



High-Performance Convolutional Network for RF-Based Drone Surveillance

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High-Performance Convolutional Network for RF-Based Drone Surveillance

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Abstract

In recent years, the increasing popularity of unmanned aerial vehicles (UAVs) has arisen from the emergence of cutting-edge technologies deployed in small and low-cost devices. With the great capability of friendly uses and wide applications for multiple purposes, amateurs drones can be piloted to effortlessly access to any geographical areas. This poses an urgently critical problem of preventing unauthorized drones from private or limited-access areas. In this paper, we propose a radio-frequency (RF)-based surveillance solution to effectively detect and classify drones, and further recognize operations by leveraging a high-performance convolutional neural network. The network, namely RF-UAVNet, is specified with grouped one-dimensional convolution to significantly reduce the network size and computational cost. Besides, a novel structure of multi-level skip-connection, for the preservation of gradient flow, incorporating with multi-level pooling, for the collection of informative deep features, is proposed to achieve high accuracy via learning efficiency improvement. In the experiments, RF-UAVNet yields the accuracy of 99.85% for drone detection, 98.53% for drone classification, and 95.33% for operation mode recognition, numbers which outperform the current state-of-the-art deep learning-based methods on DroneRF, a publicly available dataset for RF-based drone surveillance.

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Index Terms

Convolutional neural network, deep learning, drone detection, drone classification, drone surveillance.

I. INTRODUCTION

Recently, small unmanned aerial vehicles (UAVs), also known as drones, have received explosive interest unprecedentedly for numerous applications in diverse domains, which range from aerial photography to disaster management, agriculture, and communications [1]. However, this rapid development of amateur drones poses many critical threats to public security and personal privacy [2], where drones are controlled to enter restricted zones intentionally without authentication. For example, attackers might terrorize with explosives or other dangerous materials (i.e., bioterrorism or chemical) by using drones to carry out an incident. Drone surveillance has become a potential solution to effectively cope with the above-mentioned problem [3], in which drone detection and classification play an important role in many advanced anti-drone systems.

Several drone detection approaches based on radar, audio, video, and radio-frequency (RF) [4] have been developed to improve the overall performance of surveillance systems. In spite of easy installation, the radar-based detection approach suffers the challenges of detecting small drone cross-section. Although cheap and accessible, the audio-based approach is quite sensitive to noise with a limited working range (around 25-30 ft). The video-based drone identification approach undergoes several drawbacks of weak discrimination (that means, drones vs other non-drone objects) and weather constraints [5]. Compared with the others, the RF-based approach is more favourable to drone detection and classification with more substantial reliability and outstanding performance [6]. Besides easy implementation with cheap sensors, this technique has an impressive operation range (approximately 1400 ft). However, there is a minor issue of which it requires high transmit power and receiver sensitivity to achieve better detection performance [7].

In the last decade, several RF-based drone detection methods have been introduced by enabling feature extraction and machine learning (ML) algorithms to analyze RF signals. For example, a reliable drone detection method [8] was introduced to warn the presence of drones by analyzing fast Fourier transform (FFT) of RF signals over drone-controller communications. In [9], a conventional ML framework with frequency-domain feature descriptors (e.g., slope, kurtosis, and skewness) and artificial neural network classifier was studied to detect drones via WiFi

interference. In [10], nine time-domain features (including pitch, roll, yaw, thrust signals and their time derivatives) were trained with different traditional classifiers, where the random forest conducts the best performance of drone pilot identification. Lately, deep learning (DL) [11] with convolutional neural network (CNN) architecture [12], [13] has been exploited to improve the performance of drone surveillance systems. In [14], a CNN with straightforward architecture was designed to automatically learn the radio characteristics by standard one-dimensional (1D) convolutional layers. Compared with traditional classification algorithms, DL can enhance model discrimination based on its great capacity of learning features at multi-scale resolutions of signal, which performs drone detection more precisely. Nevertheless, existing DL-based methods with effortless architectures for RF-based drone surveillance suffer from two challenging issues: high implementation complexity and low accuracy. These unsolved problems impose a heavy burden on the deployments of real-time applications and reliability services with limited hardware processing in drone surveillance systems.

In this work, to overcome the above-mentioned problems, we propose an efficient CNN, namely RF-UAVNet, for high-performance RF-based drone surveillance covering drone detection, drone classification, and operation mode recognition. In particular, the network architecture is typically structured by multiple convolutional layers, in which each layer is specified by several asymmetric filters to extract local features of raw 1D signal at multi-scale resolutions (from coarse to fine). To achieve a lightweight deep model, we leverage grouped convolution instead of standard convolution to significantly reduce not only the number of network parameters but also the processing cost. Remarkably, a multi-level skip-connection is cleverly designed in an association with a multi-level pooling to improve network accuracy. The network is evaluated on the DroneRF dataset and analyzed with various hyper-parameter and architecture configurations to corroborate the efficiency of RF-UAVNet for different drone surveillance tasks. In addition, the proposed method is compared with state-of-the-art approaches with the same condition, where RF-UAVNet is more superior than other deep networks. Focusing on the novel CNN architecture, the primary contributions of this paper can be summarized as follows:

- We propose an RF-UAVNet for RF-based drone surveillance with three tasks, including drone detection, drone classification, and operation recognition. The proposed CNN employs 1D asymmetric filters of sizes 1×5 and 1×3 to calculate the local correlations of raw signal at multi-scale representations from course to fine features by structuring different convolutional layers in cascade.

- We exploit grouped convolution in RF-UAVNet to reduce the computational cost of convolutional layers by approximately the predefined number of groups. By separating the input channels into multiple groups before traversing to 1D filters to calculate features, the grouped convolution in RF-UAVNet significantly reduces the number of trainable parameters as compared to the standard convolution.
- We propose a sophisticated-designed structure, which associates multi-level skip-connection with multi-level pooling to improve learning efficiency based on the regulation of the gradient flow and the collection of the global useful features. The skip-connection with an element-wise addition operation is executed for different convolutional layers to maintain the identical gradients from the previous layers. Besides, the average pooling with different pool sizes summarizes the global features at multi-level feature representations. The meaningful features aggregated from multi-level skip-connection and multi-level pooling are forwarded to a fully connected layer for classification.
- In the experiments, we evaluate the proposed network on DroneRF, a free-accessed drone detection dataset, using the 10-fold cross-validation protocol. The performance is reported with the accuracy and the F1-score, where the proposed network achieves the overall accuracy of 99.85% for drone detection, 98.53% for drone classification, and 95.33% for operation recognition. We investigate the complexity of the network with different numbers of convolutional groups (reduces the number of parameters by around 78% and the computational cost by around 82%) and analyze the effectiveness of multi-level skip-connection associated with multi-level pooling. Furthermore, in comparison with several ML classification algorithms and existing DL-based models, our RF-UAVNet shows the superiority in different drone surveillance tasks.

The remainder of this paper is organized as follows: Section II briefly surveys the available literature of drone detection methods. In Section III, we present the proposed RF-based method for drone detection, drone classification, and operation recognition, wherein a high-performance CNN is introduced to learn RF signals. The diversified experiments with numerical results are provided in Section IV. Finally, Section V concludes this paper.

II. DRONES' DETECTION AND IDENTIFICATION - MAIN APPROACHES

The exceeding emergence of UAVs in recent years has entailed a wide variety of critical concerns on security and privacy. Consequently, numerous methods have been studied in literature

to detect drones' presence and identify their characteristics (e.g., type and operation mode), which are categorized into four main approaches, including radar-based, audio-based, video-based, and RF-based approaches. In the following, a brief explanation of these drone detection methodologies is presented.

A. Radar-Based Approach

Fundamentally, the principle of radar-based methods is the electromagnetic backscattering to measure the radar cross-section (RCS) signature, representing target reflectivity, which is used to identify an aerial object. Nevertheless, compared with aircraft, drones are more challenging for RCS-based detection because of small size and low-conductivity materials that induce low RCS. Some multistatic radar-based methods have been recently proposed to detect and track amateur drones by analysing the micro-Doppler signature. In [15], it was shown that micro-Doppler signature incorporated with time-domain analysis performed better than Doppler-shift signature in terms of the accuracy of clutter/target discrimination. The micro-Doppler signature was also adopted for drone classification in a non-cooperative drone surveillance system [16] by applying some traditional classifiers, such as support vector machine (SVM) and decision tree (DT). Recently, a multistatic passive radar with long-term evolution (LTE) downlink signals [17] was exploited to detect small UAVs. The small RCS problem caused by low flying altitude was addressed by outdoor terrestrial LTE eNodeBs and broadband communication network eNodeBs, respectively. To improve target detection performance, the authors in [18] have exploited frequency-modulated continuous wave radar with the Costas frequency coding and P4 phase coding to specify an effective LTE downlink signal feature.

B. Audio-Based Approach

These approaches can detect, classification, and localize drones based on the sounds of engine and high-speed rotating propellers captured by acoustic sensors (e.g., microphones) during the flying modes. In [19], a data-driven method was proposed to detect drones in a heavy rain environment, in which a regular ML workflow is recommended with two processing stages: feature extraction and classification. An acoustic signal of a drone is first transformed to the frequency domain using FFT and then learned by a multi-class SVM classifier. The authors in [20] improved the performance of an audio-based drone detection system with sophisticated features: linear predictive cepstral coefficients and Mel-frequency cepstral coefficients. To detect

the presence of drones and also classify acoustic signals of drones and others (such as bird, airplane, thunderstorm, rain, and wind), the work in [21] calculated the power spectral density of signal with independent component analysis for learning a classification model with the k-nearest neighbor (k-NN) algorithm. In addition, this work also proposed an effective time delay of arrival estimation scheme with the Bayesian filter and Gaussian mixture model to localize drone position accurately. Unfortunately, audio-based approaches suffer from the following limitations: sensitive to ambient noise in crowded urban areas, limited operation range, and necessity of microphone array design [7].

C. Video-Based Approach

Drone detection based on video is fundamentally a moving object detection in the field of computer vision, in which an object can be detected by analyzing visual features (e.g., color, texture, and shape) in multiple consecutive video frames. From calculating local features to automatically learning representational features, some recent approaches exploited CNNs to detect and identify drones using color camera [22] and depth camera [23]. To overcome the limited field of view, the authors in [22] recommended a moving camera in use to track small fast-flying drones. In [24], a high-performance CNN architecture was designed to boost the accuracy of drone detection while satisfying computational complexity for deployment of practical systems. The spatial attention modules integrated in the network architectures aimed to highlight small and ambiguous drones in an image for better localization and detection. Besides high sensitivity to environment illumination, these approaches have to face many challenging issues, such as occlusion and multi-object interference [25].

D. RF-Based Approach

By exploiting the intercepted RF signals between drones and ground controllers, several RF-based drone detection methods have been proposed with the advantages of day-and-night working under all weather conditions for various scenarios. A passive cost efficiency RF sensing method [26] was proposed by analyzing RF signals in the frequency domain with discrete wavelet transform and short-time fast Fourier transform to detect the presence of drones. Two vital physical characteristics of the drones (i.e., drone body shifting and body vibration) were obtained by two frequency-based techniques and then incorporated to improve the detection accuracy. In [27], the authors analyzed the intercepted RF signals at multi-resolution in the

1
2
3 wavelet domain using the discrete wavelet transform. The presence of drones was detected using
4 Naïve Bayes binary classifier incorporated with Markov models. In addition to drone detection,
5 drone classification based on WiFi and Bluetooth interference was developed by learning several
6 statistical features (denoted as RF fingerprints, including mean, standard deviation, skewness,
7 entropy, kurtosis, etc.) with some standard multi-class classifiers, such as k-NN, SVM, and RF.
8 A myriad of methodologies have investigated the traffic information of WiFi signals between
9 drone and controller for drone detection [28], [29]. In [28], the WiFi fingerprint presenting a
10 generic flow consists of several features such as the number of packets and the packet inter-
11 arrival time. The authors in [29] improved the traffic fingerprint with more descriptive statistics
12 which are optimally selected by L1-norm regularization to achieve cost efficiency. Various DL
13 architectures, such as deep neural network (DNN) and CNN, have been deployed to improve
14 the surveillance performance of RF-based systems. In [30], the authors specified three DNNs
15 for three separated tasks: drone detection, drone classification, and operation recognition. Each
16 network was principally designed with three hidden layers to learn the discrimination model
17 from FFT features of RF signals. To automatically and effectively learn features from raw data
18 (e.g., amplitude value) of RF signals, the authors in [31] designed a multi-channel convolutional
19 network (MC-CNN) with the network architecture alternately constructed by 1D convolutional
20 layers and max pooling layers. Most of the existing RF-based methods have presented a poor
21 detection performance due to the weak discrimination of conventional algorithms [27]; mean-
22 while, DL with primitive architectures cannot optimize learning efficiency. Furthermore, direction
23 of arrival estimation-based drone localization was studied in [32] by exploiting DL to process
24 single-channel RF signals. The power values of intercepted RF signals were calculated to fed
25 to a DNN having 5 layers, where the number of neurons in the input layer was identical to the
26 number of directional antenna elements.

27
28
29 In general, the drone signatures, detection range, advantages, and disadvantages of above-
30 mentioned drone detection approaches are summarized in Table I. It is worth noting that the
31 detection ranges are typically derived from the literature, that means, they might vary in the
32 practice up on the type of drones with different hardware specifications, the algorithms of
33 deployment, and the surveillance environments.

TABLE I
SUMMARY OF EXISTING DRONE DETECTION APPROACHES.

Approach	Drone signature	Detection range	Advantages	Disadvantages
Radar	RCS	≤ 3000 ft	Easy installation	Require large mono- or multi-static RF nodes
	Micro Doppler			Small RCS caused by low flying attitude Expensive device
Audio	Time-frequency feature	≤ 30 ft	Cheap sensors	Sensitive to ambient noise
			Easy implementation	Limited range
			Accessible equipment	Complex microphone array arrangement
Video	Visual features	≤ 300 ft	Easy installation Accessible equipment	Line of sight necessary
	Motion features			High resolution camera requirement Weather constraint
RF	WiFi fingerprint	≤ 1400 ft	Cheap sensors	Multipath and non-line of sight
	Time-frequency features		Easy installation	High signal-to-noise ratio necessary Vulnerable to interference

III. METHODOLOGY

A. System Model

The proposed RF-based UAV surveillance system is presented in Fig. 1. The system comprises several primary modules: drones, remote controller (a.k.a., flight control unit), RF sensing module, and processing module with database repository. Some drones for multi-purpose civil applications [33] are specified by different technical specifications, such as maximum operation range and connectivity. The remote controller, included in the drone package, sends flight commands to the target drones and also receives the response of operating status. To intercept the drone-controller communication, an RF sensing module is configured with the assumption of that the WiFi frequency is known (the operation frequency can be determined by some passive frequency scanning techniques). The intercepted RF signals can be captured by software-defined radio configurable devices and then stored in local database repository for processing hereafter.

B. RF Database Description

Regarding the aforementioned system, DroneRF [34], a large-scale RF dataset, is introduced for different tasks: drone detection, drone classification, and operation mode recognition. Four

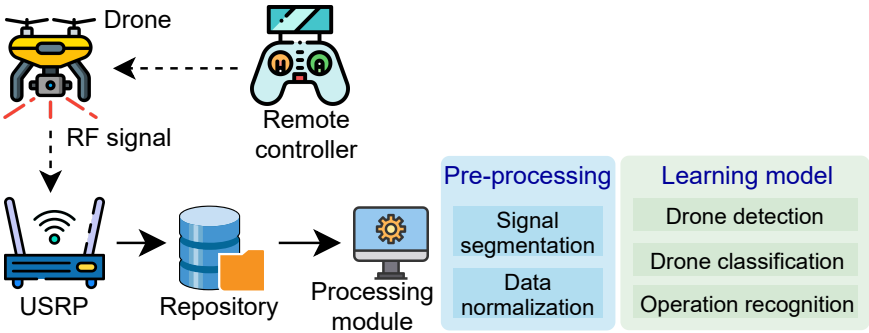


Fig. 1. System setup for RF-based UAV detection, classification, and operation mode recognition.

TABLE II
OVERALL SPECIFICATIONS OF THEE THREE DRONES UNDER CONSIDERATION.

Specifications	Parrot Bebop	Parrot AR Drone	DJI Phantom 3
Dimensions (mm)	330 × 380 × 36	517 × 517 × 127	520 × 490 × 290
Weight (g)	400	420	1216
Max. flight speed (m/s)	13	11	16
Max. wireless range (m)	300	50	1000
Connectivity	WiFi	WiFi	WiFi
Operation Frequency	2.4 GHz and 5 GHz	2.4 GHz	2.4–2.485 GHz

operation modes, denoted as modes 01 to 04, of three different drones (Parrot Bebop, Parrot AR Drone, and DJI Phantom 3, where some main specifications are listed in Table II) are considered as follows:

- Powering on and connecting to the controller.
- Hovering automatically without physical intervention and user commands.
- Flying without video recording.
- Flying with video recording.

Two USRP-2943 RF receivers with the specifications summarized in Table III are synchronized to collect the lower (L) and upper (H) half of the frequency band signals with the sampling rate of 40 MHz and the gain of 30 dB. The dataset has 227 RF signal segments, where each segment has two amplitude records (L and H bands), as shown in Fig. 2(a). It is noted that the Phantom drone is represented with one mode of drone connection (the data of other modes are corrupted in the collection procedure). Besides, the data of no drone scenario (or background activity) is

TABLE III
SPECIFICATIONS OF THE USRP-2943 RF RECEIVERS.

Specifications	Values
Number of channels	2
Range of operation frequency	1.2–6.0 GHz
Frequency step	< 1 KHz
Gain range	0–37.5 dB
Gain step	0.5 dBm
Maximum instantaneous real-time bandwidth	40 MHz
Maximum I/Q sample rate	200 MS/s
Digital-to-analog converter (DAC) resolution	14 bit

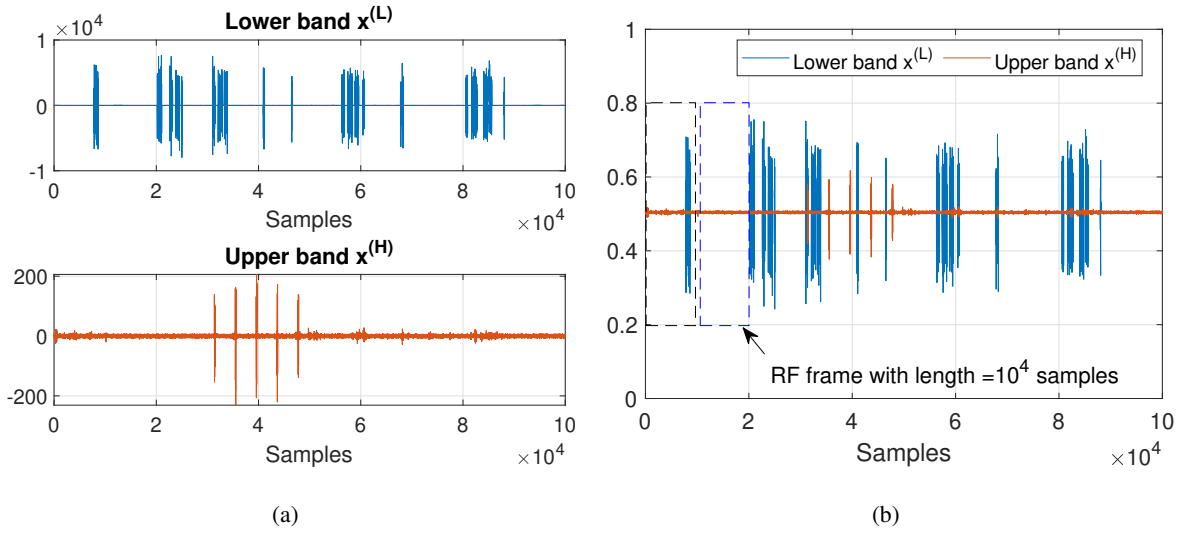


Fig. 2. Example of Bebop drone's signal: (a) the raw amplitude samples of the lower and upper bands and (b) the normalized samples with fixed-sized partition using non-overlap windowing mechanism.

recorded to serve for the detection task, which identifies whether the presence of a drone. The detailed experiments for the RF signal acquisition are presented in the original work [34].

RF data in DroneRF should be partitioned into multiple signal frames with the length of 10^4 samples (long enough to represent radio characteristic) using the non-overlap windowing mechanism [30]. Accordingly, we segment 227K signal frames, where each frame is formed in a high-dimensional array $\mathcal{S}$ with the size of $1 \times 10000 \times 2$. The information of dataset distribution is provided in Table IV. Due to the critical bias caused by two RF receivers with different RF band configurations and three drones without identical technical specifications, the amplitude

TABLE IV
DATASET DISTRIBUTION OF DRONERF [34].

Task	Class	No. segments	No. frames
Drone detection	Drone	186	186,000
	No drone	41	41,000
Drone Classification	AR	81	81,000
	Bebop	84	84,000
	Phantom	21	21,000
	No drone	41	41,000
Operation recognition	AR mode 01	21	21,000
	AR mode 02	21	21,000
	AR mode 03	21	21,000
	AR mode 04	18	18,000
	Bebop mode 01	21	21,000
	Bebop mode 02	21	21,000
	Bebop mode 03	21	21,000
	Bebop mode 04	21	21,000
	Phantom mode 01	21	21,000
	No drone	41	41,000

data needs to be normalized by scaling values to the range $[0, 1]$ as

$$\hat{x}_i^{(L)} = \frac{x_i^{(L)} - x_{\min}^{(L)}}{x_{\max}^{(L)} - x_{\min}^{(L)}}, \quad (1)$$

$$\hat{x}_i^{(H)} = \frac{x_i^{(H)} - x_{\min}^{(H)}}{x_{\max}^{(H)} - x_{\min}^{(H)}},$$

where $x_{\min}^{(L)}, x_{\max}^{(L)}, x_{\min}^{(H)}, x_{\max}^{(H)}$ denote the minimum and maximum amplitude values of the lower and upper half bands of all drones. An example of partitioning RF signal into fixed-sized frames is illustrated in Fig. 2(b).

C. RF-UAVNet: RF-based UAV Detection-Classification Convolutional Network

This part presents an efficient CNN, denoted RF-UAVNet, for low-cost and high-accuracy drone detection, drone classification, and operation recognition based on RF signals, where the

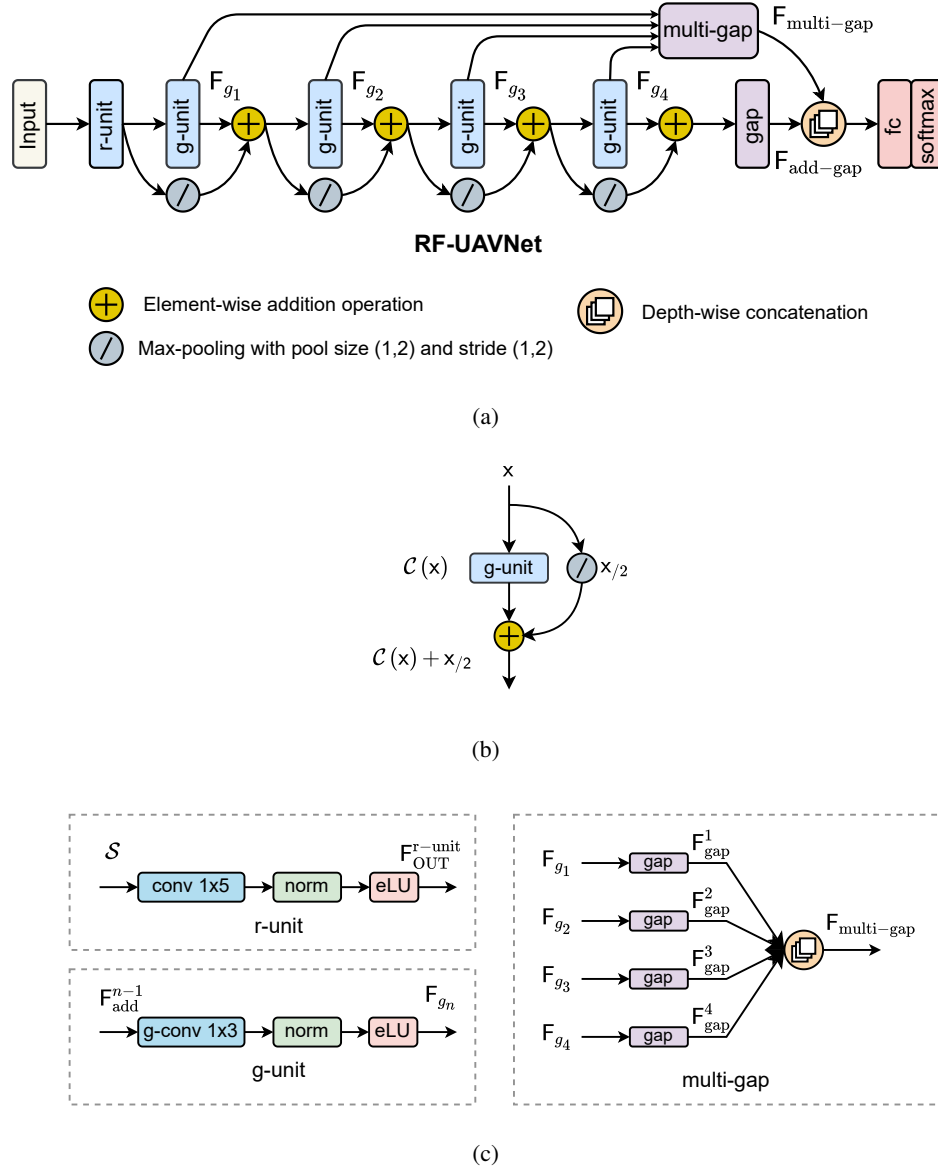


Fig. 3. RF-UAVNet: (a) overall architecture, (b) skip-connection, and (c) network components: r-unit, g-unit, and multi-gap.

overall architecture is shown in Fig. 3(a). Regarding the detailed architecture, an input layer is first specified with the size of $1 \times 10000 \times 2$ to facilitate with partitioned frames. To reduce the spatial size of feature maps to significantly save computation at deeper layers, a 1D regular convolutional (conv) layer is specified by the filters (so-called kernels) of size 1×5 with the stride of (1, 5). Notably, the number of channels for each filter in a regular conv layer is always equal to the number of channels (or the third dimension) of the input to the layer. For example, because the input having two channels connects with the conv layer directly, the number of

channels for each filter is two. The conv layer is followed by a batch normalization (norm) layer and an exponential linear unit (eLU) activation layer, where this layer group is denoted r-unit in Fig. 3(b). Fundamentally, the convolution between a kernel with weights w_i and an input map u_i at any specific coordinate (x, y) in the spatial domain is formulated as

$$z_{x,y} = \sum_i w_i u_i + b, \quad (2)$$

where b is the scalar bias. The volume convolution of K channels (a.k.a., the depth size of the input) is the sum of K resulting convolution scalar z . The norm layer performs value normalization to its input with the mean μ_B and variance σ_B^2 calculated for each mini-batch and input channel as

$$\hat{z}_{x,y} = \frac{z_{x,y} - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}}, \quad (3)$$

where $\epsilon = 10^{-5}$ is the constant scalar to prevent the numerical uncertainty caused by a very small variance. To possibly process the inputs with zero mean and non-optimal variance passing to the layer that follows the norm layer, the norm layer then shifts and scales the normalized values by

$$v_{x,y} = \kappa \hat{z}_{x,y} + \xi, \quad (4)$$

where κ and ξ denote, respectively, the scale factor and offset parameters that are updated during network training. Once the training process is finished, the norm layer calculates the mean and variance over the whole training set. In the testing process to predict the class of a new input, the norm layer uses the trained mean and variance instead of the mean and variance of a mini-batch to normalize the activation. In a CNN for classification, batch normalization is typically used between conv and eLU layers to accelerate the training of networks and reduce the sensitivity to network initialization [35]. The eLU layer then performs nonlinearity by outputting the identical value on positive inputs and the exponential operation result on negative inputs, i.e.,

$$o_{x,y} = \text{eLU}(v_{x,y}) = \begin{cases} v_{x,y}, & \text{if } v_{x,y} \geq 0, \\ \alpha (e^{v_{x,y}} - 1), & \text{if } v_{x,y} < 0, \end{cases} \quad (5)$$

where $\alpha = 1$ is the nonlinear parameter. Compared with the rectified linear unit commonly developed in several convolutional networks, eLU can achieve the learning convergence faster with a higher accuracy in some classification tasks [36]. With 64 filters of size 1 and the stride of $(1, 5)$ specified for the conv layer, the width dimension of output maps reduces five times

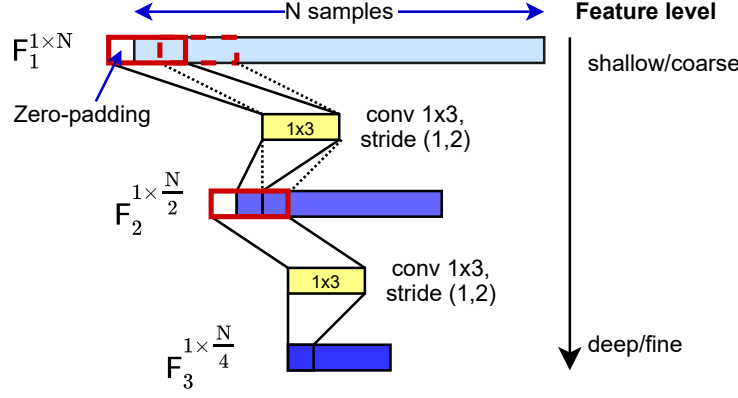


Fig. 4. Example of the 1D convolution operation to extract local features at multi-scale resolutions from coarse to fine.

and the depth dimension is identical to the number of filters. At this point, with an input signal frame $\mathcal{S} \in \mathbb{R}^{1 \times 10000 \times 2}$, we determine the output of the r -unit $\mathbf{F}_{\text{OUT}}^{r-\text{unit}} \in \mathbb{R}^{1 \times 2000 \times 64}$.

As the primary modules of RF-UAVNet to extract the local correlations of amplitude samples at multi-scale feature representations (as illustrated in Fig. 4), four processing units (denoted g-unit) are arranged in cascade throughout the network architecture in Fig. 3(a). In each g-unit, a grouped convolutional (g-conv) layer with the filters of size 1×3 and the stride of $(1, 2)$ is followed by an eLU layer (see Fig. 3(c)). Regarding the grouped convolution, the filters are separated into different groups for processing, where each group will serve as standard convolution for a certain number of input maps. Compared with the standard convolution, the grouped one reduces the number of parameters and computational cost significantly further as the number of non-overlapping filter groups increases, while the learning efficiency is mostly maintained with the sparse relationship among filters [37]. With G groups, the complexity of g-conv is $\frac{D_{\text{in}} \times D_{\text{out}}}{G}$, which is lower than that of conv by G times [38], where D_{in} and D_{out} , respectively, denote the depth size of input and output. An example of grouped convolution with two groups is illustrated in Fig. 5, where the first group (yellow) convolves with the first half of the input and the second group (red) convolves with the second half. The number of filter groups must evenly divide the number of channels of the input layer. The number of channels in the output of g-conv layer is $\text{numGroups} \times \text{numFiltersPerGroup}$, where numGroups and $\text{numFiltersPerGroup}$, respectively, denote the number of groups and the number of filters for each group. In this work, we specify the g-conv layers by eight groups with each group having

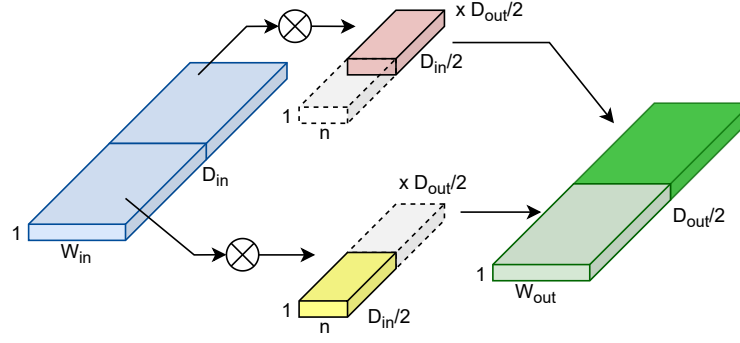


Fig. 5. Example of the grouped 1D convolution with two filter groups.

eight filters to always produce 64 feature maps at the output of g-conv layers. We denote by $\mathbf{F}_{g_{i=1,\dots,4}} \in \mathbb{R}^{1 \times W_{g_i} \times 64}$ with $W_{g_i} = \{1000, 500, 250, 125\}$ the outputs of g-units, where W_{g_i} is the horizontal size of the output maps resulted in the g-conv layer with the stride of (1, 2) in the i -th g-unit.

To increase the overall system accuracy via well-handling gradient flow in RF-UAVNet, we study a multi-level skip-connection. Inspired by [39], skip-connection associates the outputs of different g-units by adopting element-wise addition operation. Given the output of r-unit $\mathbf{F}_{OUT}^{r-unit}$, the procedure of the first skip-connection can be formulated with an element-wise addition (add) layer as

$$\begin{aligned} \mathbf{F}_{add}^1 &= \mathcal{F}(\mathbf{F}_{OUT}^{r-unit}), \\ &= \mathbf{F}_{p1} + \mathbf{F}_{g1}, \end{aligned} \quad (6)$$

where $\mathbf{F}_{p1} = \mathcal{P}_{max}(\mathbf{F}_{OUT}^{r-unit})$ and $\mathbf{F}_{g1} = \mathcal{C}(\mathbf{F}_{OUT}^{r-unit})$; $\mathcal{P}_{max}$ denotes the max-pooling operation with the pool size of (1, 2) and the stride of (1, 2); $\mathcal{C}$ denotes the united operation of grouped convolution, batch normalization, and eLU activation. The max pooling (maxpool) layer is adopted to pick on the most salient features [40] and to align the output to spatial size of $\mathbf{F}_{g1}$. The element-wise addition operation in (6) is done by an addition (add) layer, where all inputs to this layer must have the same dimension. For example, by downsampling $\mathbf{F}_{OUT}^{r-unit}$ in the horizontal dimension with the stride of (1, 2), the output of maxpool layer $\mathbf{F}_{p1} \in \mathbb{R}^{1 \times 1000 \times 64}$ has the same dimension with $\mathbf{F}_{g1} \in \mathbb{R}^{1 \times W_{g1} \times 64}$. In neural networks, the vanishing gradient problem usually occurs when the gradient elements (the partial derivatives with respect to the network parameters) become exponentially small; hence, the update of the parameters with the gradient is almost negligible. Based on the principle of residual block [39], the skip-connection allows the

network to learn the residual $\mathcal{C}(\cdot)$ instead of learning the true output $\mathcal{F}(\cdot)$ to nearly maintain the gradient flow. To put it simply, the attenuated gradient caused by activation functions is consolidated with the identity information by skipping layers. The structure of skip-connection deployed in RF-UAVNet is generally described in Fig. 3(b). For a multi-level mechanism, the n -th skip-connection with $n \geq 2$, in general, can be expressed as follows:

$$\begin{aligned}\mathbf{F}_{\text{add}}^n &= \mathcal{F}(\mathbf{F}_{\text{add}}^{n-1}), \\ &= \mathbf{F}_{p_{n-1}} + \mathbf{F}_{g_{n-1}}, \\ &= \mathcal{P}_{\text{max}}(\mathbf{F}_{\text{add}}^{n-1}) + \mathcal{C}(\mathbf{F}_{\text{add}}^{n-1}).\end{aligned}\quad (7)$$

For the purpose of tracking the dimension of feature maps along with the backbone of RF-UAVnet in the corporation with multi-level skip-connection, the output halves the width dimension of feature maps for each time of passing the input through a skip-connection. In particular, we obtain $\mathcal{F} : \mathbf{F}_{\text{add}}^1 \in \mathbb{R}^{1 \times W_{g1} \times 64} \leftarrow \mathbf{F}_{\text{OUT}}^{\text{r-unit}}$ for the first skip-connection and $\mathcal{F} : \mathbf{F}_{\text{add}}^n \in \mathbb{R}^{1 \times W_{gn} \times 64} \leftarrow \mathbf{F}_{\text{add}}^{n-1} \in \mathbb{R}^{1 \times W_{g_{n-1}} \times 64}$ for the n -th skip-connection with $n \geq 2$, where W_{gn} is the width of output maps. It is worth noting that the element-wise addition layer does not change the feature dimension. The proposed multi-level skip-connection allows the gradient information to be mostly maintained at different feature levels. Consequently, the vanishing/exploding gradient problem can be prevented effectively when the network goes deeper.

A global average pooling (gap) layer is configured to calculate the mean of each feature map, which is usually located before fully connected (fc) layers for network size reduction and overfitting alleviation with no learnable parameters. For example, the gap layer transforms the dimensions from $1 \times W \times D$ to $1 \times 1 \times D$ by averaging across with the pool size of $(1, W)$ for every maps along the channel dimension. To collect the meaningful knowledge from preceding layers, we propose a multi-level global average pooling (multi-gap) module as described in Fig. 3(c). The outputs of different g-unit $\mathbf{F}_{gi}$ are first forwarded to the gap layers individually to obtain the global features

$$\mathbf{F}_{\text{gap}}^i = \mathcal{P}_{\text{avg}}(\mathbf{F}_{gi}), \text{ with } i = 1, \dots, 4, \quad (8)$$

where $\mathbf{F}_{\text{gap}}^i \in \mathbb{R}^{1 \times 1 \times 64}$ and $\mathcal{P}_{\text{avg}}$ denotes the average pooling operation with different pool sizes. The resulting global features are then stacked by a depth-wise concatenation (concat) layer as

$$\mathbf{F}_{\text{multi-gap}} = \mathcal{D}(\mathbf{F}_{\text{gap}}^1, \mathbf{F}_{\text{gap}}^2, \mathbf{F}_{\text{gap}}^3, \mathbf{F}_{\text{gap}}^4), \quad (9)$$

where $\mathcal{D}$ denotes the concatenation operation. It is noted that a concat layer takes the inputs having the same height and width and aggregates them along the depth (or channel) dimension. The output $\mathbf{F}_{\text{multi-gap}} \in \mathbb{R}^{1 \times 1 \times 256}$ has the number of channels (or depth size) that is the sum of those of all inputs.

At the end of network, the output of multi-gap $\mathbf{F}_{\text{multi-gap}}$ combines with the output of the last skip-connection $\mathbf{F}_{\text{add-gap}} = \mathcal{P}_{\text{avg}}(\mathbf{F}_{\text{add}}^4)$ by a concat layer, see Fig. 3(a), as follows:

$$\mathbf{F}_{\text{final}} = \mathcal{D}(\mathbf{F}_{\text{multi-gap}}, \mathbf{F}_{\text{add-gap}}), \quad (10)$$

where $\mathbf{F}_{\text{final}} \in \mathbb{R}^{1 \times 1 \times 320}$, which contains the synthetic information of signal characteristics at multiple scales, is flattened into a single vector before feeding to the fc layer defined by k neurons and followed by a softmax layer. The number of neuron in fc layer is identical to the number of classes in a given dataset for a particular task. Regarding different drone surveillance tasks studied in this work, we use the same CNN architecture with a minor change in the fc layer. Particularly, RF-UAVNet is with $k = 2$ for drone detection, $k = 4$ for drone classification, and $k = 10$ for operation recognition (see Table IV). The detailed network configurations are further provided in Table V. The network is finalized with a softmax layer that performs a normalized exponential function to transform the scores $\mathbf{s} = \{s_j\}_{j=1}^k$ deduced by the fc layer to the class probability distribution

$$p_j = \frac{e^{s_j}}{\sum_{i=1}^k e^{s_i}}, \quad (11)$$

where p_j is the j -th class probability of an input data with $0 \leq p_j \leq 1$ and $\sum_{i=1}^k p_i = 1$. The softmax function can be considered the multi-class generalization of the logistic sigmoid function. The network assigns each input to one of k mutually exclusive classes based on the output of softmax function and computes the cross entropy loss for multi-classes classification as follows:

$$\mathcal{L} = - \sum_{i=1}^N \sum_{j=1}^k \nu_{ij} \ln(v_{ij}), \quad (12)$$

where N is the number of training signal frames, ν_{ij} denotes the ground-truth of the i -th input signal associated with the j -th class, and v_{ij} denotes the output resulted by the network for the class j of the input signal i .

The configurations of network training process are given as follows: the optimizer is the stochastic gradient descent with momentum, the momentum factor is 0.95, the L_2 regularization factor is 0.0001, the maximum number of epochs for training is 90, the initial learning rate is

TABLE V
RF-UAVNET CONFIGURATION (TASK 01: DRONE DETECTION, TASK 02: DRONE CLASSIFICATION, AND TASK 03:
OPERATION RECOGNITION).

Layer	Network description			Output Size
	Task 01	Task 02	Task 03	
input	RF signal frame			$1 \times 10000 \times 2$
r-unit	64 filters 1×5 , stride (1, 5)			$1 \times 2000 \times 64$
g-unit	8 groups of 8 filters 1×3 , stride (1, 2)			$1 \times 1000 \times 64$
g-unit	8 groups of 8 filters 1×3 , stride (1, 2)			$1 \times 500 \times 64$
g-unit	8 groups of 8 filters 1×3 , stride (1, 2)			$1 \times 250 \times 64$
g-unit	8 groups of 8 filters 1×3 , stride (1, 2)			$1 \times 125 \times 64$
gap	pool size (1, 125)			$1 \times 1 \times 64$
multi-gap	4 gap layers + depth-wise concat			$1 \times 1 \times 256$
concat	depth-wise concat			$1 \times 1 \times 320$
fc	$k = 2$	$k = 4$	$k = 10$	$1 \times 1 \times k$

0.01 (dropping to 0.001 after 45 epochs for a better training convergence), and the mini-batch size is 128. The network is evaluated on a platform using 3.70-GHz CPUs, 32GB RAM, and a single NVIDIA GTX 1080Ti GPU.

IV. EXPERIMENTAL RESULTS AND DISCUSSIONS

This section evaluates the performance of RF-UAVNet for drone surveillance using the 10-fold cross-validation protocol [30]. Concretely, the DroneRF dataset is randomly partitioned into ten non-overlapping folds, where one fold is used as the validation set for testing, and the remainder is used to train the network. This process is then repeated ten times to evaluate the whole dataset, and the overall performance is reported by averaging the performance of all folds.

Specifically, four principal experiments along with insightful discussions are delivered as follows:

- In the first experiment, we benchmark the performance of RF-UAVNet for three drone surveillance tasks (i.e., drone detection, drone classification, and operation recognition denoted by tasks 01-03) on the DroneRF dataset.

- The second experiment evaluates the parameter sensitivity of RF-UAVNet to the overall system performance with different numbers of filter groups in g-conv layers.
- The third experiment is an ablation study, in which RF-UAVNet is compared with its variants to demonstrate the advantages of grouped convolution and skip-connection incorporated with multi-gap.
- The last one compares RF-UAVNet with other state-of-the-art DL-based approaches, which also exploit DNN and CNN models for drone surveillance.

The performance of drone detection and classification is measured using the accuracy and F1-score metrics, which are described as follows:

$$\text{accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (13)$$

$$\text{F1 - score} = \frac{2TP}{2TP + FP + FN}, \quad (14)$$

where TP, TN, FP, and FN are the true positive, true negative, false positive, and false negative, respectively.

A. Model Performance

In the first experiment, we evaluate the performance of RF-UAVNet on the DroneRF dataset for different tasks, where the numerical results of confusion matrices are shown in Fig. 6. For drone detection with the result given in Fig. 6(a), RF-UAVNet reaches the overall correct identification rate of 99.85%. For drone classification, RF-UAVNet yields the overall accuracy of 98.55%, where Phantom presents the worst classification rate of 96.41%. Phantom and Bebop are confused with AR by around 2.80% and 2.04%, respectively. For operation recognition which is more challenging than drone detection and classification with more classes for discrimination, the network achieves the overall accuracy of 95.33%. From the confusion matrix presented in Fig. 6(c), Bebop mode 04 (lying with video recording) presents the worst recognition rate of 88.37%, which is mostly confused with Bebop mode 03 (flying without video recording) by approximately 8.63%. Interestingly, some high confusions are found between different modes of the Bebop drone, for instance, mode 01 (powering on and connecting) and mode 02 (hovering) are misclassified as mode 03 with the error rate of 4.31% and 5.74%, respectively.

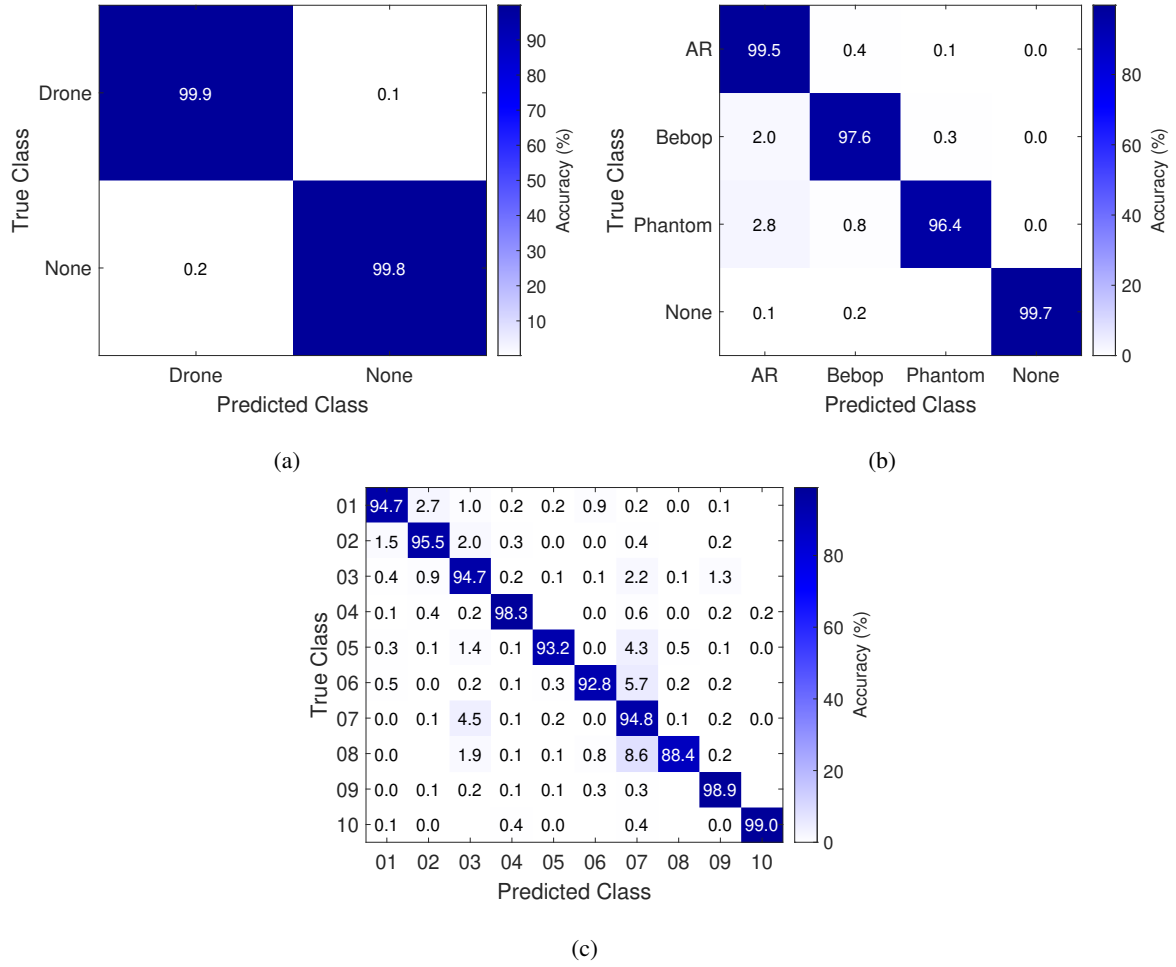


Fig. 6. Confusion matrices: (a) drone detection, (b) drone classification, and (c) operation mode recognition, where classes 01-04 are four modes of the AR drone, classes 05-08 are four modes of the Bebop drone, class 09 denotes the Phantom mode 01, and class 10 denotes RF background activity without drone detection.

B. Parameter Sensitivity

This experiment investigates the relationship between the overall performance and the network complexity with various numbers of filter groups (numGroups) defined in g-conv layers. The complexity is evaluated by the network size, which is measured by the number of trainable parameters, and the computational cost, which is estimated by the number of floating point operations (FLOPs). In Fig. 7, we report the results of three tasks with numGroups = {2, 4, 6, 8, 16, 32}, where the number of filters in each group is automatically determined to produce the same number of feature maps. In general, the cost efficiency of RF-UAVNet gains along with the increment of numGroups, whereas the performance of three drone surveillance tasks gets degradation. For instance, from Figs. 7(a) and 7(b), the F1-score of drone detection and classification reduces

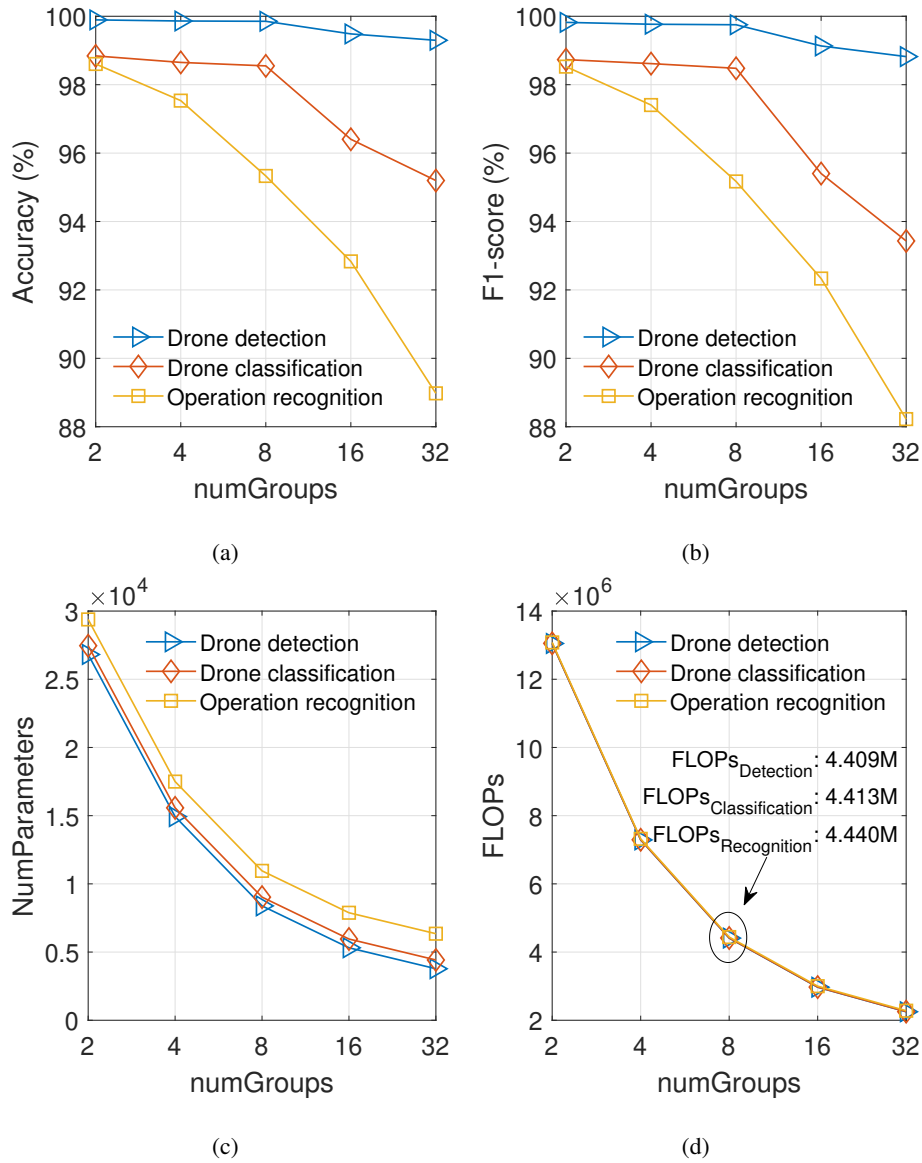


Fig. 7. Performance and complexity of RF-UAVNet with various number of filter groups (numGroups) regarding: (a) accuracy, (b) F1-score, (c) number of parameters, and (d) FLOPs.

by around 0.01% and 0.14%, respectively, when we increase numGroups from 4 to 8 groups. However, when we increase numGroups from 8 to 16 groups, the F1-score decreases severely by 0.62% for drone detection and 3.08% for drone classification. For operation recognition, the performance deterioration is nearly identical in each time of doubling numGroups (approximately the accuracy of 2.41% and the F1-score of 2.57%). Concerning network complexity, the number of parameters and FLOPs reduce significantly for small numbers of filter groups. Because of the same architecture configuration of r-unit and g-units to extract features, the size

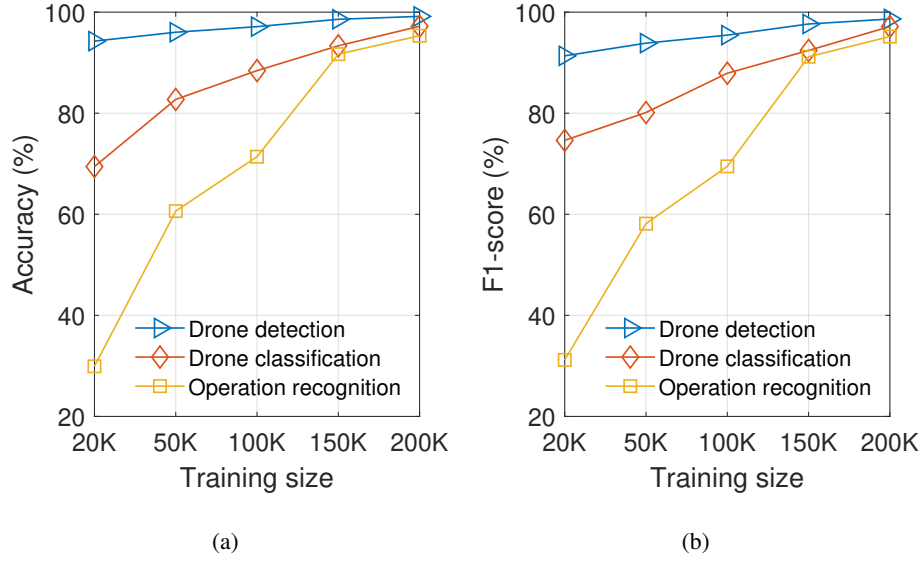


Fig. 8. Performance of RF-UAVNet with different sizes of the training set.

difference of RF-UAVNets for different tasks derives from the numbers of weights and biases to densely associate all neurons in the concat layer with k neurons in the fc layers. With $k = 10$, the size of RF-UAVNet for operation recognition is bigger than those of the networks for drone detection and drone classification as shown in Fig. 7(c). Notably, the computation of RF-UAVNet mostly is for feature calculation in convolutional layers, while the computational cost consumed in the fc layer is insignificant. This explains a very small gap of FLOPs for three different tasks in Fig. 7(d). Based on the experimental benchmark, we specify $\text{numGroups} = 8$ to achieve a comfortable balance between performance and complexity.

C. Impact of Training Data Size

In this experiment, we study the impact of the size of training set on the network performance for different surveillance tasks. To this end, we split the whole dataset with 227K RF signals randomly into the training set to learn classification model and the testing test to benchmark the accuracy and F1-score of the trained model. In particular, we vary the total number of RF signals for training in a range $[20K, 200K]$, where the remainder is for testing. The procedure of randomly splitting data, training model, and estimating performance is repeated many times to report the average results in Fig. 8. In general, the performance is improved further as the training size increases. For drone detection, the overall accuracy and F1-score increase by 4.90% and 7.30% when the size of training set is enlarged by 10 times, i.e., from 20K to 200K signals.

TABLE VI
PERFORMANCE COMPARISON BETWEEN RF-UAVNET AND ITS CUT-OFF VARIANTS.
(COMPLEXITY IS REPORTED WITH RF-UAVNET FOR OPERATION RECOGNITION).

Network	Strategy				Accuracy (%)			F1-score (%)			Approx.Complexity	
	conv	g-conv	skip-conn	multi-gap	Task 01	Task 02	Task 03	Task 01	Task 02	Task 03	No.params	FLOPs
Net-A	✓				99.70	92.72	83.98	99.49	90.98	82.55	50K	24.5M
Net-B		✓			99.50	91.14	83.21	99.15	90.43	81.86	8.4K	4.4M
Net-C		✓	✓		99.71	95.77	92.62	99.51	93.75	92.06	8.4K	4.4M
RF-UAVNet		✓	✓	✓	99.85	98.55	95.33	99.75	98.48	95.06	11K	4.4M

The performance improvement is more pronounced in the drone classification benchmark, where the average accuracy and F1-score increase by 6.94% and 5.63%, respectively, for each time of augmenting the size of training set. Notably, the performance of operation recognition increases remarkably along with the growth of the training set’s size. Obviously, the impact of the training size is more forceful to a more challenging task. Concretely, when the number of RF signals for training model increases from 20K to 50K, the overall accuracy and F1-score raise about 30.72% and 26.97%, respectively. The performance improvement becomes minor from 150K to 200K training signals.

D. Ablation Study

The third experiment evaluates the effectiveness of grouped convolution incorporated with multi-layer skip-connection and multi-gap mechanisms, which are exploited in RF-UAVNet. Specifically, we compare RF-UAVNet with three cut-off variants, denoted by Net-A, Net-B, and Net-C, in terms of accuracy and complexity. The detailed architectures of these CNNs are given as follows (or being briefly summarized in Table VI):

- Net-A: an alternative architecture of RF-UAVNet using standard conv layers without the incorporated structure of skip-connection and multi-gap.
- Net-B: an alternative architecture of RF-UAVNet using g-conv layers without skip-connection and multi-gap (or Net-A with g-conv).
- Net-C: an alternative architecture of RF-UAVNet using g-conv layers and skip-connection without multi-gap (or Net-B with skip-connection).

It is noted that all networks are evaluated with the same training configuration. Compared with Net-A which uses standard conv layers, Net-B with g-conv layers remarkably reduces the network size (from 50K to 8.4K parameters) and computational cost (from 24.5 MFLOPs to 4.4 MFLOPs, where MFLOPs denotes megaFLOPs), while it suffers a trivial reduction of performance. Net-C, which is upgraded from Net-B with the multi-layer skip-connection, achieves a performance improvement in terms of accuracy and F1-score without the increment of the number of parameters and FLOPs. By leveraging a sophisticated-designed architecture, wherein g-conv layers associated with multi-layer skip-connection and multi-gap mechanisms, RF-UAVNet successfully obtains a twofold objective: low-complexity and high accuracy. Concretely, RF-UAVNet is approximately 80% lower than Net-A in terms of complexity and effectively performs drone classification and operation recognition better than Net-B by around 7.41 – 12.12%. Notably, RF-UAVNet is bigger than Net-B and Net-C in terms of the network size because the multi-gap structure for multi-level feature maps aggregation increases the number of parameters in the fc layer.

E. Method Comparison

In the last experiment, we compare the proposed CNN-based method with the conventional methods (which adopt feature extraction and various traditional classification algorithms) and the innovative DL-based methods on the DroneRF dataset. It is worth noting that all the comparison methods are evaluated using the same evaluation protocol (i.e., 10-fold cross-validation). In the conventional methods, we calculate the statistical features (e.g., mean, absolute mean, variance, standard deviation, kurtosis, skewness, etc) [27] of amplitude samples of RF signal, including the lower and upper bands. For each signal, a set of 30 features (15 per each band) is organized into a vector to input various ML algorithms with some major hyper-parameters as follows:

- Linear discriminant analysis (LDA): the discriminant type of diagonal linear, the linear coefficient threshold $\delta = 0$, and the factor of regularization $\gamma = 0.5$.
- k-NN: the number of neighbors $n = 10$ and the distance metric of Mahalanobis.
- SVM: the coding design is one-vs-one with the radial basis function kernel and the sequential minimal optimization.
- DT: the maximum number of decision splits $k = 100$ and the split criterion of maximum deviance reduction (a.k.a., cross entropy).

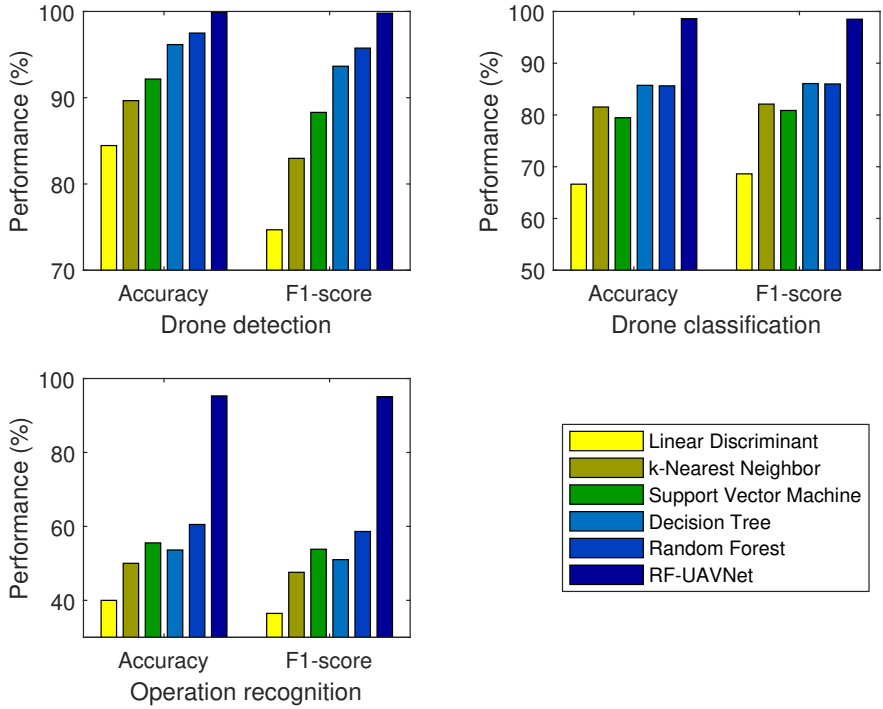


Fig. 9. Performance comparison with ML algorithms.

- RF: the maximum number of decision splits $k = 100$, the number of learners $l = 60$, and the learner type of decision tree.

Based on the results in Fig. 9, LDA reports the worst performance of all tasks among different traditional classifiers, for example, the overall accuracy and F1-score regarding the operation recognition task are approximately 40.0% and 36.5%. SVM, a widely used multi-class classification algorithm, performs better than k-NN and DT in drone detection and operation recognition but showing a worse accuracy and F1-score in drone classification. RF, which is built by aggregating multiple DT learners, can overcome the individual tree correlation problem (an essential drawback of regular DT) by sub-sampling features, thus enhancing classification performance. RF significantly outperforms DT in the operation recognition benchmark with the greater accuracy and F1-score of approximately 6.9% and 7.61%, respectively. The proposed DL model forcefully defeats all testing ML algorithms for various drone surveillance tasks. Concretely, RF-UAVNet is better than RF (a.k.a., the second-best classification algorithm of this experiment) by the overall accuracy and F1-score of 2.41% and 4.05% for drone detection, and 12.4% and 12.5% for drone classification. The remarkable performance improvement is yielded in the operation

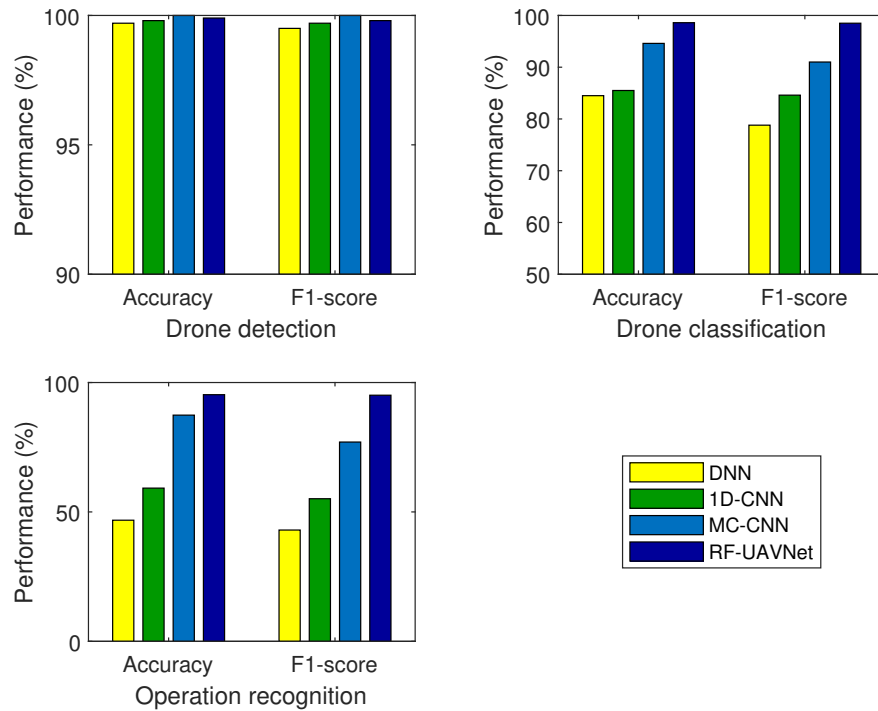


Fig. 10. Performance comparison with DL models.

recognition benchmark with greater accuracy and F1-score of 34.8% and 36.5%, respectively. Obviously, with the ability to learn features automatically at different levels, from coarse to fine, our proposed CNN is more superior than several common classifiers.

Additionally, the RF-UAVNet is compared with some recent DL models deployed for drone surveillance, including DNN [30], 1D-CNN [14], and MC-CNN [31]. The numerical results are presented in Fig. 10 in terms of the overall accuracy and F1-score. In general, the performance gap between the comparison methods in drone detection is trivial (around 0.1 – 0.5%), whereas the gaps in drone classification and operation recognition are significant. DNN, which is simply designed with three hidden layers, reports the worst performance, especially with the operation recognition task with the overall accuracy of 46.8% and the F1-score of 43.0%. By leveraging convolutional architectures, 1D-CNN with multiple 1D conv layers associated with average pooling layers in a cascade structure performs better than DNN in drone classification and operation recognition. Compared with DNN, 1D-CNN improves accuracy and F1-score by up to 12.4% and 12.1%, respectively. With a depth-wise concatenation layer to selectively aggregate features in multiple channels, MC-CNN reaches the drone classification accuracy of 94.6% and the operation recognition accuracy of 87.4%, which significantly outperforms DNN by

9.1 – 28.2%. Despite being worse than MC-CNN by a tiny gap of drone detection (around 0.1 – 0.2%), RF-UAVNet, which effectively incorporates the multi-level skip-connection and multi-gap mechanisms, performs drone classification and operation recognition more precisely with the higher accuracy of 4.0% and 7.9% and with the greater F1-score of 7.5% and 18.1%, respectively. Accordingly, the proposed CNN has demonstrated the superiority in terms of accuracy against other existing deep models for drone surveillance.

V. CONCLUSION AND DISCUSSIONS

In this paper, we proposed a cost-efficient and high-accuracy CNN, namely RF-UAVNet, for drone detection, drone classification, and operation recognition, which was employed in the implementation of RF-based drone surveillance systems. RF-UAVNet was characterized by grouped convolutional layers to reduce the network size and computational cost significantly, where each g-conv layer was specified by asymmetric 1D kernels to capture the temporal correlations as local features of RF signals. Remarkably, to increase the overall accuracy of system, we designed the multi-level skip-connection and multi-gap mechanisms to, respectively, prevent the vanishing gradient effectively and collect the useful global features at different signal resolutions. In the experiments, we analyzed RF-UAVNet exhaustively with different architecture configurations on the DroneRF dataset to demonstrate the effectiveness of grouped convolution to reduce complexity and multi-level skip-connection incorporated with multi-gap to improve accuracy. RF-UAVNet achieved very high accuracy for different drone surveillance tasks (approximately 99.9% for drone detection, 98.6% for drone classification, and 95.3% for operation recognition) with low-complexity (11K parameters and 4.4 MFLOPs), which could be a favorable approach for onboard deployment in portable anti-drone systems. Furthermore, the proposed deep network considerably outperformed traditional ML models and state-of-the-art DL models for drone surveillance in terms of the accuracy and F1-score.

The proposed RF-based drone surveillance approach has studied a sophisticated-designed deep network with comprehensive performance analysis in terms of accuracy and complexity. Consequently, our up-to-date research could play principal literature and inspire more works. Besides, we found some challenging issues which will be possibly handled in our future work.

- The intra-class and inter-class variations caused by silent duration in RF signals conduct the performance degradation of multi-class classification.

- The proposed network achieves high accuracy for 10-class classification of operation recognition, but balancing accuracy and complexity becomes more difficult as the number of classes increases.

Drones are being exploited progressively as an important part of many wireless systems for wide-ranging applications, from civil to the military, hence studying drone surveillance with performance analysis is essential. Accordingly, some open research problems can be taken into consideration.

- *Multi-modality surveillance systems*: Combining different drone surveillance approaches (e.g., radar, audio, video, and RF) for multi-level information fusion (i.e., data-level, feature-level, and decision-level) can improve performance besides effectively handling the drawbacks of each approach performed separately.
- *Data synchronization*: One of the most challenging and fundamental issues of data acquisition and management is synchronization between different signal sources, where the device specifications and parameter configurations are disparate. Data synchronization allows updating changes automatically to maintain system consistency.
- *Multiple drones surveillance*: Most of the existing drone surveillance approaches have been studied for single drone detection. Identifying numerous drones' simultaneous presence in a region of interest necessitates data association to distinguish the data, acquired from heterogeneous sensors, of corresponding drones.

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